



Subnano-sized silicon anode via crystal growth inhibition mechanism and its application in a prototype battery pack

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Due to the large volume variation of high-capacity alloy-based anodes during cycling, it is desirable to use small anode particles for an extended battery cycle life. However, it is still challenging to realize subnano-sized particles (<1 nm). Here we show a growth inhibition mechanism that prevents continuous enlargement of size immediately after nucleation during chemical vapour deposition. The growth inhibition is successfully applied to the synthesis of silicon, thereby yielding subnano-sized (<1 nm) silicon embedded in a highly stable dual matrix composed of carbon and silicon carbide. Ethylene not only functions as a silicon growth inhibitor, thereby slowing the growth of nucleated silicon via Si–C bond formation, but also acts as a source to create the dual matrix. The subnano-sized silicon anode enhances the cycling stability (Coulombic efficiency reaching 99.96% over 50 cycles). Finally, the practical application of the fabricated energy storage system (103.2 kWh) containing 110 Ah full-cells with 91% capacity retention for 2,875 cycles and a calendar life of 97.6% for 1 year is demonstrated.

Replacing conventional graphite anodes with high-capacity materials is the most promising way to achieve higher energy density lithium-ion batteries¹. Silicon (Si), which reacts with lithium via an alloying reaction, has been considered as a feasible candidate because of its high specific capacity (3,592 mAh g^{−1})^{2–5}. However, Si suffers from severe structural degradation and instability of the solid–electrolyte interphase (SEI) caused by substantial volumetric fluctuations during (de)lithiation^{6,7}. These can not only bring about particle fracture and loss of electrical contact with the electrode system but also mechanically disturb the stable formation of the SEI layer and continuously expose the new surface of the material to the electrolyte.

To mitigate these issues, numerous pioneering works over the past decade have proven that decreasing the feature size to the nanoscale allows Si to tolerate the huge dimensional strains without fragmentation and to provide excellent electrochemical behaviours^{8–10}. Furthermore, according to previously reported computational studies, subnano-sized Si (approximately 1 nm) leads to much less strain^{11–13}. Nonetheless, previous studies on Si-based anodes have focused on managing morphological properties (nanosheets, nanowires, nanotubes, hollow structures and yolk–shell structures) with sizes over several nanometres^{14–16}. To the best of our knowledge, the properties of subnano-sized Si anodes have been reported only through simulation because most preceding synthetic approaches, including alloying metals, reduction of silicon oxide, sol–gel, chemical vapour deposition and mechanical milling methods, have limitations in realizing subnano-sized Si (refs. ^{17–19}). Other challenges include high surface area, low tapping

density and air instability (excessive oxidation) when decreasing the Si size^{18,20}. The high surface area causes severe side reactions with the electrolyte while consuming confined Li⁺ ions. The low tapping density of these materials leads to a sparsely packed electrode, which makes it difficult to achieve robust interparticle electronic and ionic pathways, high mass loading and finally high volumetric energy. In addition, Si inevitably has a native oxidation layer (SiO₂) with a thickness of 2–3 nm in an air atmosphere, but it is negligible for micrometre-sized Si (refs. ^{21,22}). However, for subnano-sized Si, native oxides of even a few nanometres could occupy a considerably large portion of the total weight, resulting in lowering the specific capacity, hindering ionic and electronic conductivity and inducing an irreversible phase.

To address these concerns, incorporation of Si into a composite with other elements has been considered as a rational strategy for small-sized Si. Various carbonaceous materials, including graphite, graphene and amorphous carbon, have been adopted as a buffer matrix for Si composites to alleviate volume changes, block direct exposure of the Si surface to the electrolyte and provide electrically conductive channels^{23–29}. However, they are not robust enough to withstand severe volume fluctuations and are prone to breakage during long-term cycles^{30–33}. Alternatively, Si suboxide (SiO_x, $x \leq 1$), consisting of a nano-Si domain embedded in a SiO₂ matrix, demonstrates good comprehensive performance and commercial feasibility because of its tiny Si nanocrystals (4–8 nm) and sturdy oxide matrix covering the Si domains conformally. Notwithstanding these advantages, since the oxygen element of SiO_x produces fatal irreversible phases such as Li₂O and Li₄SiO₄, the use of SiO_x over a

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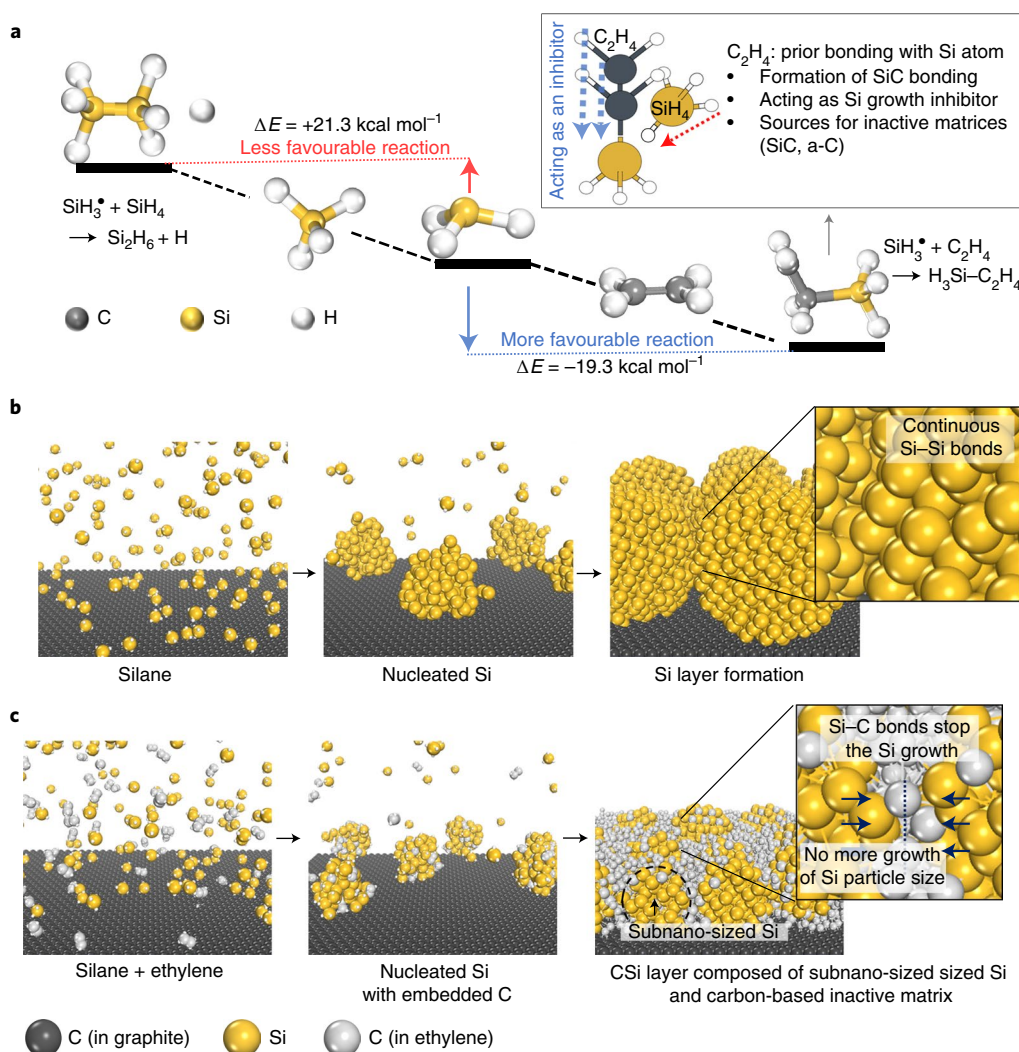


Fig. 1 | Simulation regarding the behaviour of Si growth inhibitor. a, Calculated heat of reaction (ΔE) for the reaction of the SiH_3 radical with ethylene (C_2H_4) and silane (SiH_4). Inset: a schematic illustration of the behaviour of ethylene gas acting as a Si growth inhibitor and preventing continuous Si-Si bonding through prior bonding with Si atoms. **b,c**, Schematic illustrations showing the thermal decomposition of monosilane (**b**) and the mixture of monosilane and ethylene (**c**).

certain amount induces large irreversible capacity losses^{34,35}. Hence, SiO_x has been commercialized only on a limited scale ($\leq 4 \text{ wt}\%$). On the basis of these facts, we concluded that gradually downsizing Si would lead to more-enhanced characteristics because of improved damage tolerance. Moreover, introducing a robust and Li-inactive matrix could effectively help to address the formidable issues of subnano-sized Si.

The nucleation of Si continues to grow and generates a Si layer during the thermal decomposition of monosilane gas. Hence, we expected that subnano-sized Si would be generated if its growth was obstructed immediately after the nucleation. Through theoretical simulations, we found a growth inhibition mechanism in which the formation of Si-Si bonds could be prevented by ethylene while forming multiple Si-C bonds (Fig. 1a), which eventually led to the formation of subnano-sized Si clusters during thermal decomposition. Based on this constructive hypothesis and calculation, we synthesized the subnano-sized Si ($< 1 \text{ nm}$) embedded in a stable dual matrix of silicon carbide and amorphous carbon (SiC/a-C matrix) in a layer structure (denoted the CSi layer) via chemical vapour deposition by using complex gases (SiH_4 and C_2H_4). The CSi layer covered carbonaceous supporting frameworks (controllable high capacity over $1,262 \text{ mAh g}^{-1}$ and Coulombic efficiency

(CE) of 90.0%), which showed excellent cycling stability by preserving its morphological integrity without any cracks or pulverization. Consequently, a battery pack (107 kWh) containing 110 Ah full-cells was fabricated to verify the practical application of this anode, which demonstrated cycling stability (91% for 2,875 cycles) and a good calendar life (97.6% for 365 days).

Computational study of Si growth inhibition mechanism

To verify the hypothesis regarding the role of ethylene as a Si growth inhibitor, the thermal decomposition pathway of the gas mixture (monosilane and ethylene) was computationally studied via density functional theory (DFT) calculations (details in Methods, Supplementary Note 1, Supplementary Figs. 1 and 2 and Supplementary Table 1). We found that the SiH_3 radical that comes from the decomposition of SiH_4 preferably reacts with ethylene rather than with monosilane. Consequently, the existence of ethylene interrupts the continuous growth of nucleated Si through the formation of the Si-C bond, which results in the formation of subnano-sized Si enclosed in SiC/a-C matrices, unlike the case of the monosilane decomposition without ethylene (Fig. 1b,c). To theoretically validate the Si growth dynamics during thermal decomposition, we performed reactive molecular dynamics (reax-MD) simulations

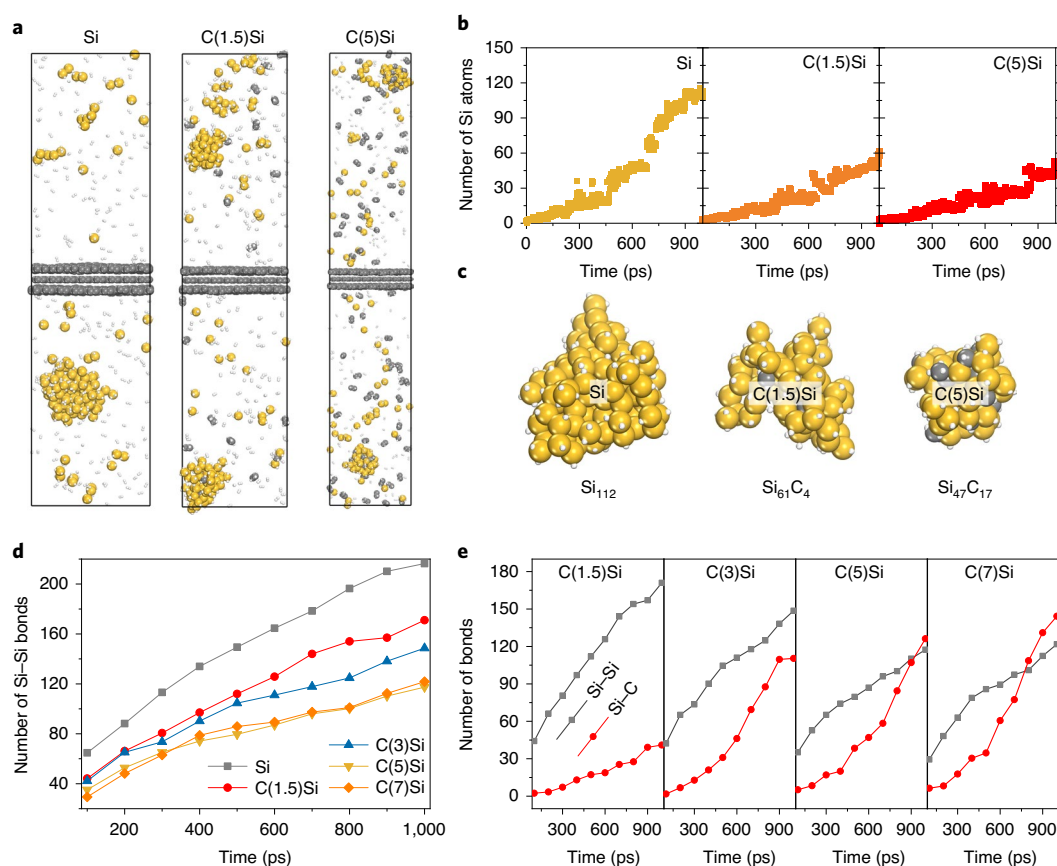


Fig. 2 | Investigating the growth of subnano-sized Si through MD. **a**, Reax-MD simulation snapshots of the Si, C(1.5)Si and C(5)Si systems after 1,000 ps. The C(*x*)Si slab model (*x* = 0, 1.5, 5) consists of three layers of graphene and a mixture of silane (159 molecules) and ethylene (0, 26 and 91 molecules for the three systems, respectively). Carbon, hydrogen and silicon atoms are coloured in grey, white and yellow, respectively. **b**, The number of Si atoms in the largest atom cluster over time. The yellow, orange and red indicate Si, C(1.5)Si and C(5)Si systems, respectively. **c**, The largest atom cluster in the last configuration of the Si, C(1.5)Si and C(5)Si systems. The atomic composition of Si and C atoms in each atom cluster is Si₁₁₂, Si₆₁C₄ and Si₄₇C₁₇ for Si, C(1.5)Si and C(5)Si, respectively. **d**, The number of Si-Si bonds over time. **e**, Comparison of the number of Si-Si and Si-C bonds during the reax-MD simulation.

(details in Methods, Supplementary Note 1, Supplementary Figs. 3 and 4 and Supplementary Table 2). The ratios of SiH₄ to C₂H₄ were 10:0, 10:1.5 and 10:5, and were denoted as Si, C(1.5)Si and C(5)Si, respectively. As the ratio of ethylene to monosilane increased from pristine Si to C(5)Si, the growth rate of the atom cluster decreased substantially (Fig. 2a,b and Supplementary Figs. 5 and 6), resulting in the smallest particle in the C(5)Si system in terms of the number of Si atoms (Fig. 2c). Furthermore, the number of Si-Si bonds greatly decreased as ethylene increased, whereas the number of Si-C bonds increased (Fig. 2d,e). This is because carbon atoms adsorbed at the surface of Si atoms prevent the continuous formation of Si-Si bonds. These atoms therefore suppress the growth of Si clusters, finally leading to the formation of subnano-sized Si and the SiC/a-C matrices. From the above comprehensive results, we conclude that Si growth can be efficiently managed using ethylene acting as a growth inhibitor during the thermal decomposition of the precursor gas.

Based on these calculations, we synthesized subnano-sized Si embedded into SiC/a-C matrices on a graphite substrate via thermal decomposition of SiH₄ and C₂H₄; the Methods section describes the detailed experimental work. Scanning electron microscopy (SEM) images of the pristine Si layer and C(5)Si layer on the graphite are shown in Fig. 3a,b (different magnified SEM images in Supplementary Fig. 7). Figure 3c,d shows high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) images of the cross-sectional pristine Si and C(5)Si layer. The thickness of the C(5)Si layer (22 nm) was slightly

greater than that of the pristine Si layer (15 nm) because of the existence of SiC/a-C in the C(5)Si layer.

We conducted electron energy loss spectroscopy (EELS) with transmission electron microscopy (TEM) analysis to investigate the bonding in the CSi layer. In the carbon K-edge region, σ^* (~293 eV) and π^* (~286 eV) peaks were clearly observed in the C(5)Si layer, and these peaks indicate the existence of carbon bonding in the layer (Fig. 3e). On the other hand, the pristine Si layer had no peaks for carbon bonding in the carbon K-edge region. In addition, we found the Si L₂₃-edge peak (~99.2 eV) assigned to the Si bonding in the pristine Si layer, whereas the C(5)Si layer provided a shifted Si L₂₃-edge peak at about 102.6 eV (Fig. 3f). Such a shifted peak implies the coexistence of Si-Si and Si-C bonding in the C(5)Si layer. According to the energy-dispersive X-ray spectroscopy (EDS) analysis (Supplementary Fig. 8), the C(5)Si layer consists of 36.8% and 63.1% of carbon and silicon elements, respectively. X-ray photoelectron spectroscopy (XPS) also indicates Si-Si and Si-C bonding in the C(5)Si layer with a relatively limited amount of Si-O bonding compared to the Si layer, as shown in Fig. 3g,h (detailed interpretation in Supplementary Note 2 and Supplementary Fig. 9). The quantitative composition of each bonding was calculated based on the XPS and TEM-EDS analyses (Fig. 3i and Supplementary Note 3).

Tuning Si crystallite size through the growth inhibitor

We investigated the effects of the Si growth inhibitor on the Si crystallite size by tuning the ratio of the growth inhibitor from pristine

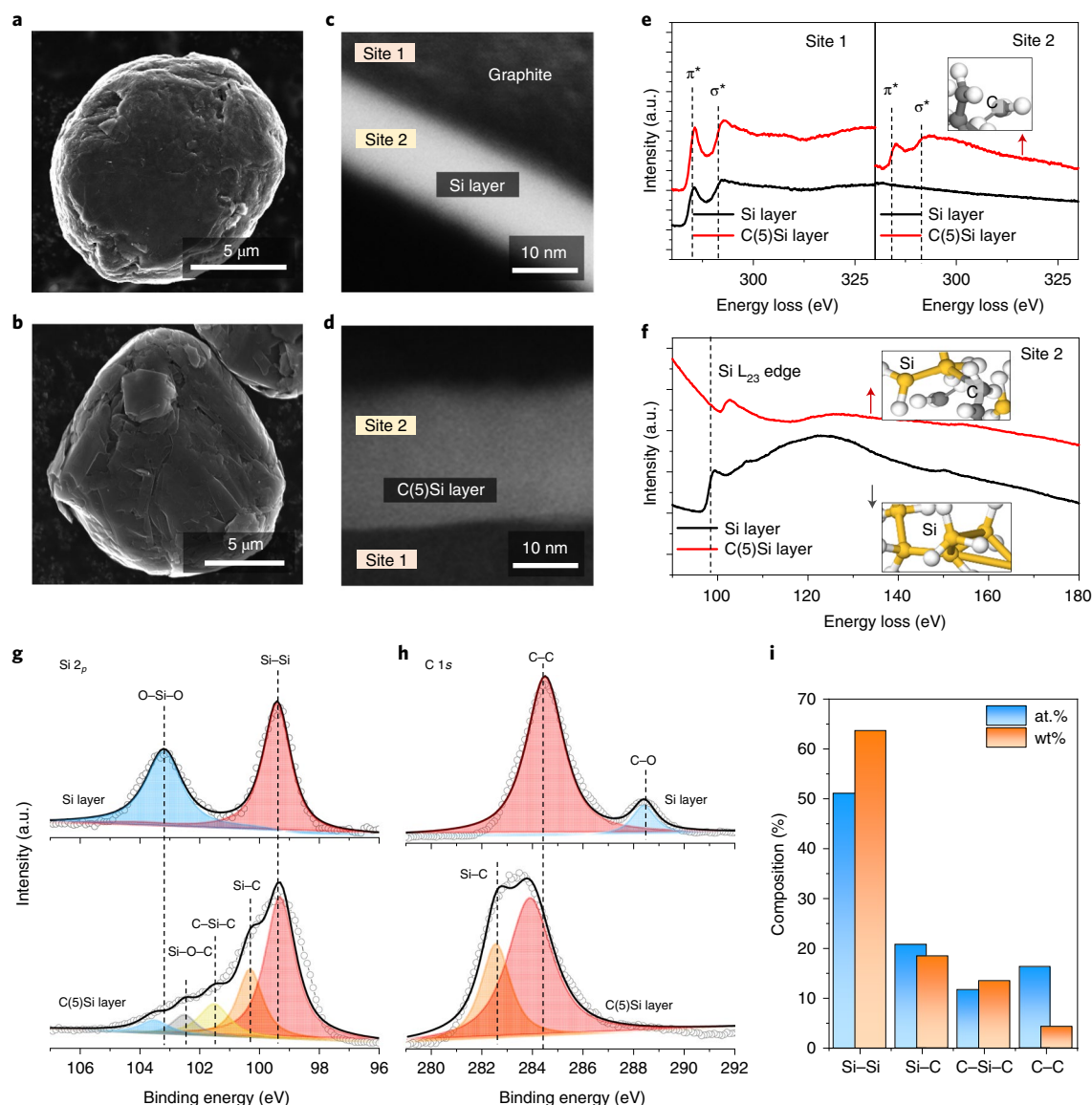


Fig. 3 | Chemical characterization of composition and bonding in CSi layer. **a,b**, SEM images of the top view of Si-G (**a**) and C(5)Si-G (**b**). Each experiment was repeated at least three times independently to obtain similar results. **c,d**, Cross-sectional HAADF-STEM images showing Si layer (**c**) and C(5)Si layer (**d**) on the graphite. **e,f**, EELS spectra of the carbon K edge (**e**) and Si L₂₃ edge (**f**) corresponding to Site 1 and Site 2 in the HAADF-STEM images, with insets illustrating the chemical bonds expected to be in each layer. **g,h**, The high-resolution XPS of Si 2p (**g**) and C 1s (**h**) for the Si and C(5) Si layers. Red, orange, yellow, grey and blue indicate Si-Si, Si-C, C-Si-C, Si-O-C, and O-Si-O bonds, respectively. The black line is the fitting results of XPS and the grey circles indicate original data of XPS. **i**, Bonding composition (weight and atomic percentages) of Si-Si, Si-C, C-Si-C and C-C, calculated based on the XPS and TEM-EDS analysis.

Si to a C(7)Si layer. To identify the crystallographic behaviour of Si in the growth-inhibited CSi layer, X-ray diffraction analysis was performed after crystallization through a heat treatment (Supplementary Note 4 and Supplementary Figs. 10–13 demonstrating the characteristics of Si and the CSi layer before annealing). In addition, we applied the CSi layer on the amorphous carbon substrate, rather than graphite, to avoid the dilution of the Si-related peaks by the graphitic carbon peaks with relatively high intensity. The X-ray diffraction pattern of the CSi layer coated on graphite is presented in Supplementary Fig. 14. In Fig. 4a, the pristine Si layer shows highly intense and sharp peaks assigned to crystallite Si without any trace of the SiC pattern. By contrast, the CSi layer provided very weak and broad Si and SiC peaks compared to those of the pristine Si layer. Note that the Si peak intensities and their full width at half maximum decrease with an increase in the ratio of

crystal growth inhibitor. These X-ray diffraction patterns indicated the limit of crystallization of Si by the introduction of the growth inhibitor, and its crystallite size became finer as the growth inhibitor content increased. In addition, the X-ray diffraction results revealed that the emergence of the SiC phase was derived from the growth inhibition mechanism, which acted as an inactive buffer matrix for Si without the formation of irreversible Li compounds. To investigate further, we applied the Scherrer equation to calculate the crystallite size of Si in each sample (Fig. 4b). To remove aberrations coming from instrumental broadening during measurement of X-ray diffraction^{36,37}, the Si standard material peak (111) was used to calculate crystallite size (Supplementary Fig. 15). The size of the Si crystal in the pristine Si layer is about 40 nm, but the CSi layer had a highly reduced crystallite size from 3.8 nm to 0.87 nm due to the hindering of the crystal growth. Notably, the C(5)Si layer has subnano-sized

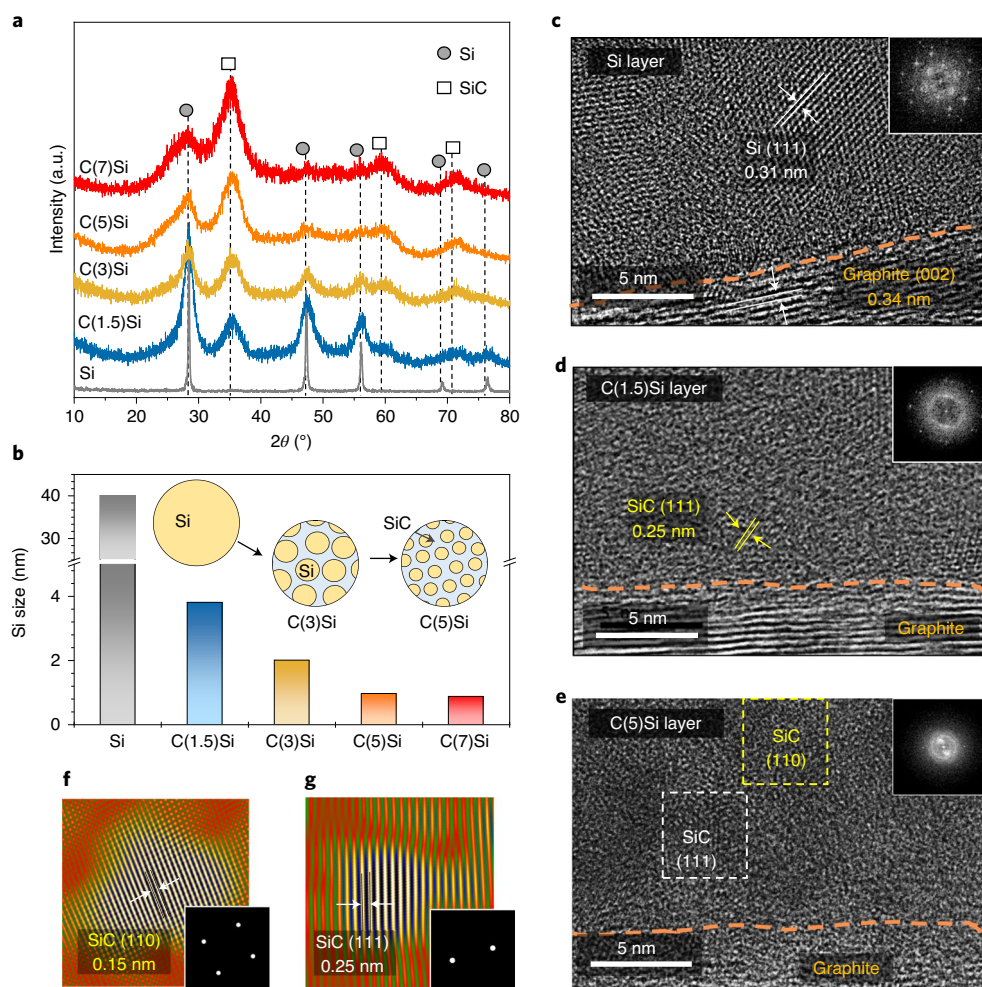


Fig. 4 | Investigation of crystallite size in CSi layer. **a**, X-ray diffraction pattern of Si, C(1.5)Si, C(3)Si, C(5)Si and C(7)Si layers coated on the amorphous carbon substrate. **b**, Crystallite size of Si calculated by the Scherrer equation and schematic inset showing the change of Si size as the ratio of ethylene increases during gas decomposition. **c–e**, High-resolution TEM images of Si-G (**c**), C(1.5)Si-G (**d**) and C(5)Si-G (**e**) with inset FFT images. The orange dotted curves indicate the boundaries between graphite and the Si or CSi layer, and the parallel lines show the d -spacing of each material. **f, g**, Inverse FFT images of SiC (110) and (111) in the C(5)Si layer corresponding with the white square (**f**) and yellow square (**g**), respectively, in the TEM images.

Si of approximately 0.96 nm, and these subnano-sized Si crystals, because of their structural reversibility, are greatly advantageous for repeated (de)alloying with Li ions.

High-resolution TEM images of each sample with fast Fourier transform (FFT) analysis are shown in Fig. 4c–e. The pristine Si layer provided a large and clear Si crystallite phase with high crystallinity (Fig. 4c). However, as the amount of crystal growth inhibitor increased, the size of the Si crystals substantially decreased in the C(1.5)Si layer with the emergence of the SiC phase (Fig. 4d). Additionally, it is very difficult to distinguish the Si crystallite phase in the C(5)Si layer (Fig. 4e) because of its subnanometre size (inverse FFT images of SiC in Fig. 4f,g). These TEM results agreed well with the X-ray diffraction data and the calculation of the crystallite size. In this regard, we confirmed that the size of the Si crystals can be effectively tuned by the crystal growth inhibitor.

Electrochemical characterization of CSi layer

To verify the favourable effect of the CSi on battery performance, electrochemical characterization was conducted (Fig. 5; the details of cell preparation and electrochemical measurements are provided in the Methods and Supplementary Table 3). The general properties (ionic conductivity and cyclic voltammetry curve) of the

electrolyte we used are shown in Supplementary Fig. 16. We obtained the first-cycle voltage (V) profiles of each sample with differential capacity (dQ/dV) analysis (Fig. 5a).

It is well known that crystallite Si (c-Si) undergoes amorphization during electrochemical lithiation while forming the Li_xSi phase (a- Li_xSi). However, when the content of lithium in a- Li_xSi reaches a critical value ($x=3.75$), a- Li_xSi is converted to the crystallite $\text{Li}_{15}\text{Si}_4$ phase (c- $\text{Li}_{15}\text{Si}_4$), which gradually grows at potentials lower than 50 mV (ref. 38). This recrystallization phenomenon causes additional internal stresses associated with phase transition and growth. Eventually, recrystallization leads to particle fracture, which is detrimental to cycling, and is particularly reactive to the electrolyte and the parasitic reaction that removes active Li ions. For these reasons, suppressing c- $\text{Li}_{15}\text{Si}_4$ is considered an important criterion for determining its performance.

Interestingly, in Fig. 5a, the voltage plateau at approximately 450 mV, which corresponds to the phase transition of c- $\text{Li}_{15}\text{Si}_4$, is notably diminished by the introduction of the crystal growth inhibitor. The pristine Si layer on graphite (pSi-G) clearly had a characteristic peak at 450 mV, but the C(1.5)Si layer on graphite (C(1.5)Si-G) showed a highly reduced peak in the same region. Then, upon testing C(5)Si-G, the characteristic peak vanished. This phenomenon

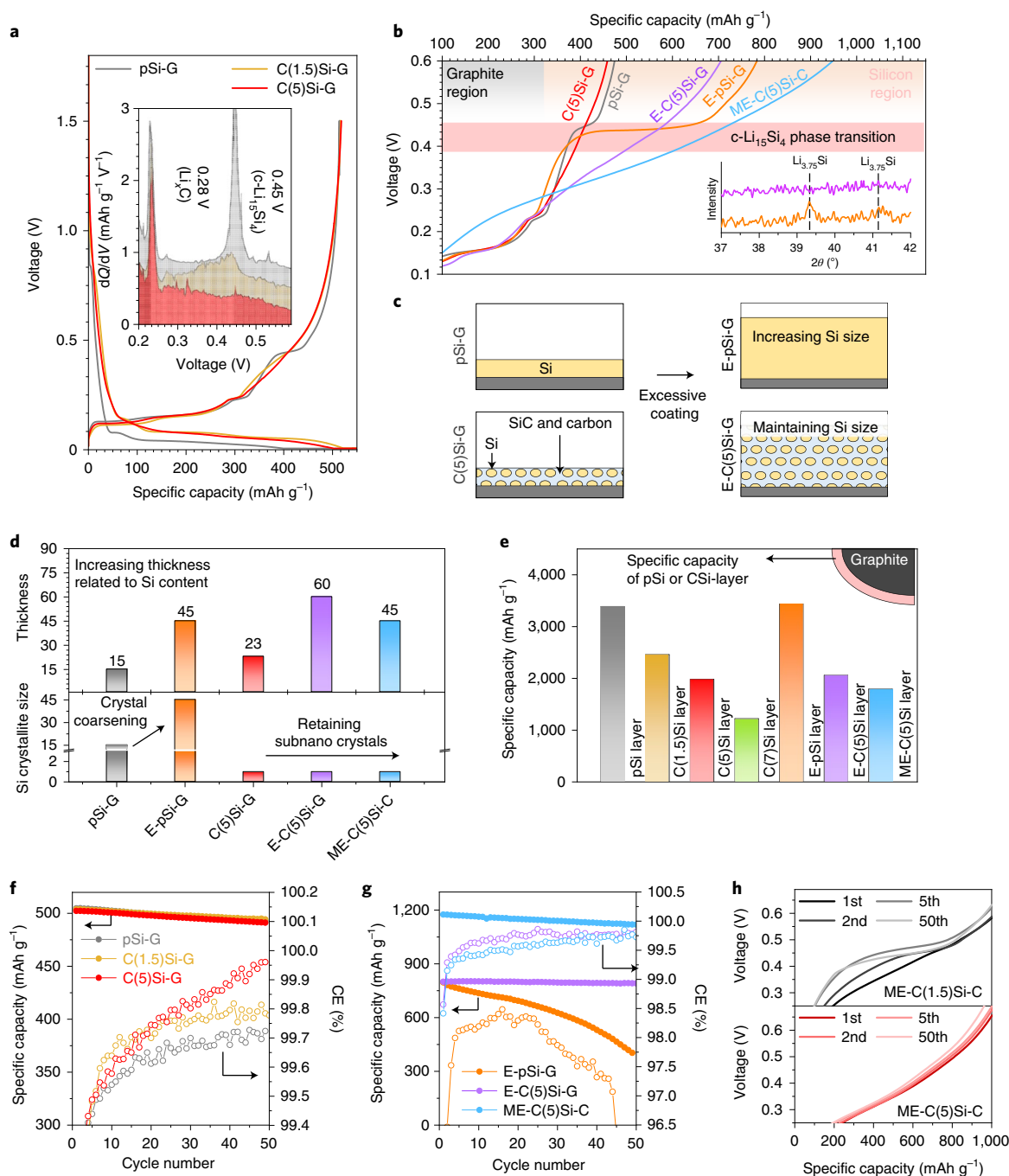


Fig. 5 | Electrochemical characterization of various anodes. a, Voltage profiles of pSi-G, C(1.5)Si-G and C(5)Si-G with an inset of the dQ/dV plot in half-cells at the first cycle. **b**, Magnified voltage profile of pSi-G, C(5)Si-G, E-pSi-G, E-C(5)Si-G and ME-C(5)Si-C and inset showing X-ray diffraction analysis of the $\text{c-Li}_{15}\text{Si}_4$ phase after lithiation (purple and orange lines indicate E-C(5)Si-G and E-pSi-G, respectively; the dashed lines indicate peaks assigned to crystallite $\text{Li}_{3.75}\text{Si}$ phase). The specific capacity derived from graphite and Si are indicated by the graphite region (grey) and Si region (orange), respectively. A voltage plateau of ~ 450 mV indicating the phase transition of $\text{c-Li}_{15}\text{Si}_4$ is marked in pink on the graph. **c**, Schematic showing properties regarding normal and excessive pSi and C(5)Si coating. **d**, Change of coating thickness depending on the pSi and C(5)Si content and its Si crystallite size (coating thickness is based on the magnified cross-sectional SEM images). The coating thickness of ME-C(5)Si-C does not increase as much as the Si content increase because the high surface area of the carbon substrate provides enough space to be coated. **e**, Pure gravimetric capacity of each layer (three different synthesized samples were measured and the intermediate value was plotted). **f, g**, Discharging capacity retentions of pSi-G, C(1.5)Si-G, C(5)Si-G, E-pSi-G, E-C(5)Si-G and ME-C(5)Si-C when measured at 0.5 C, and the CE per cycle. **h**, The dQ/dV plots of ME-C(1.5)Si-C and ME-C(5)Si-C at different cycle numbers: the 1st, 2nd, 5th and 50th.

originated from both the effect of subnano-sized Si and robust SiC/a-C dual matrices, which can effectively suppress the evolution of the unfavourable $\text{c-Li}_{15}\text{Si}_4$ phase by the matrix-induced external force. More interestingly, even with excessive (approximately three

times) coating of pSi and C(5)Si layers on the graphite—denoted as E-pSi-G and E-C(5)Si-G—and more excessive (approximately seven times) coating of the CSi layer on the carbon (denoted ME-C(5)Si-C; its SEM images are shown in Supplementary Fig. 17), the

recrystallization phenomenon was still completely suppressed. On the other hand, a much larger characteristic peak for $c\text{-Li}_{15}\text{Si}_4$ was observed in E-pSi-G than that of pSi-G (Fig. 5b and Supplementary Fig. 18). These results indicate that the subnanometre size of Si is maintained and the dual matrix still works well in the C(5)Si regardless of the coating amount (Fig. 5c). Figure 5d shows that as the Si content increased, the size of the Si crystal also increased with crystal coarsening, and the Si coating thickness increased. However, the subnano-sized Si crystals in the C(5)Si layer were retained regardless of their coating thickness (Supplementary Fig. 19 shows X-ray diffraction analysis of E-pSi-G and E-C(5)Si-G, and the measurement of coating thickness was based on the magnified cross-sectional SEM images in Supplementary Fig. 20). These results indicate that it is possible to increase the Si amount without any side effects of the growth of Si size, which has been a severe limitation for high specific capacity via a chemical vapour deposition process¹⁹. More surprisingly, these subnano-sized Si crystals and the dual matrix do not have an adverse effect on the initial CE (E-C(5)Si-G (92.8%) and E-pSi-G (93.1%) have similar values). This is because SiC/a-C dual matrices have intrinsically high reversibility, unlike the SiO_x anode accompanying the irreversible phase formation. Furthermore, when specific capacities of each the pSi and CSi layer are calculated and plotted, shown in Fig. 5e (Supplementary Note 5, Supplementary Figs. 21 and 22 and Supplementary Table 4), the specific capacity of the C(5)Si layer is similar regardless of the thickness of the C(5) Si layers in C(5)Si-G, E-C(5)Si-G and ME-C(5)Si-C. The specific capacity and weight ratio of each component in the anodes is shown in Supplementary Fig. 23.

In the cycling test, we observed that the CE of all samples of pSi-G, C(1.5)Si-G and C(5)Si-G increased to 99.5% after four or five cycles, which is similar to our previously reported Si-graphite hybrids¹⁹. However, the CE of C(5)Si-G reached 99.96% at the 50th cycle, whereas the CE was 99.72% and 99.78% for pSi-G and C(1.5)Si-G, respectively (Fig. 5f). Considering that the average CE reaching at least 99.96% is imperative for a capacity retention ratio of over 80% during 500 cycles (calculation in Supplementary Fig. 24), these results indicate that C(5)Si-G is qualified for long-term cycling performance.

To further confirm the stability of subnano-sized Si, we investigated E-pSi-G, E-C(5)Si-G and ME-C(5)Si-C (Fig. 5g). As the Si content increased, E-pSi-G exhibited a poor cycle life with unstable cycling efficiency. However, E-C(5)Si-G still provided good cycling stability with gradually increasing cycling efficiency. We also compared ME-C(5)Si-C with commercial graphite to confirm the electrochemical stability of the C(5)Si layer by excluding the effects of the well-known stable nature of the graphite. Even though the specific capacity of ME-C(5)Si-C ($1,262.3 \text{ mAh g}^{-1}$) is much higher than that of graphite (362.0 mAh g^{-1}), it shows comparable cycling stability (Supplementary Fig. 25 shows detailed electrochemical tests including the half-cell and full-cell tests of ME-C(5)Si-C, ME-C(5)Si-C blending with graphite and graphite). This result indicates that the C(5)Si layer still demonstrates excellent cycling stability, even with approximately seven times more coating content than that of C(5)Si-G.

To substantiate the superiority of the C(5)Si, we investigated the differential capacity (dQ/dV) plots characterizing the delithiation of ME-C(1.5)Si-C and ME-C(5)Si-C during cycling (Fig. 5h). A voltage plateau of approximately 450 mV (phase transition of $c\text{-Li}_{15}\text{Si}_4$) continues to emerge after the first cycle in the case of ME-C(1.5) Si-C, implying that the Si size continues to increase through the coalescence of lithiated Si with adjacent Si upon cycling. By contrast, ME-C(5)Si-C has no characteristic peak at 450 mV, which means that the subnano-sized Si, with a size of less than 1 nm, retains its morphological integrity while inhibiting the coalescence of Si upon repeated lithiation and delithiation processes. These results provided a reason why realizing the subnano-sized Si is imperative for stable high-capacity Si anodes with long-term cycling stability.

Full-cell evaluation and post-cycling analysis of the CSi layer

To verify the practical feasibility of the CSi technology, we prepared a 1.1 Ah pouch-type full-cell. Supplementary Fig. 26a provides the electrochemical information of the cathode, and magnified voltage profiles exhibiting the open-circuit voltage of the anode and cathode in the half-cell and full-cell are shown in Supplementary Figs. 27 and 28. The C(5)Si-G demonstrated an excellent cycling performance of over 80% capacity retention over 500 cycles under industrial evaluation conditions (Fig. 6a). By contrast, the capacity retention of pSi-G was 57.6% (voltage profile is shown in Supplementary Fig. 26b, and the cell design and electrochemical estimation (an areal capacity of $>3.3 \text{ mAh cm}^{-2}$) are given in Supplementary Table 5 and Supplementary Fig. 26c, respectively).

In addition, we evaluated other critical requirements for practical use, such as the cycling stability and electrode volume expansion at a high temperature (45°C), at which the electrode degradation is accelerated (Supplementary Fig. 29). Even at a high temperature (45°C), C(5)Si-G provided a superior capacity retention ratio (82.3%) over 200 cycles and a low electrode expansion of less than 35% during cycling. C(5)Si-G also demonstrated an outstanding rate capability of 79.6% capacity retention at 5 C (Supplementary Fig. 30 for voltage profile at a different current of discharge).

A post-cycling analysis of the 1 Ah stacking cell was conducted to investigate the advantageous effect of subnano-sized Si with SiC/a-C dual matrices for long-term cycling. Supplementary Fig. 31 shows the cross-sectional SEM images of pSi-G and C(5)Si-G before cycling. Both pSi-G and C(5)Si-G exhibited a well-attached coating layer on the graphite. However, after 500 cycles, pSi-G showed a highly degraded and detached from the graphite Si layer (Fig. 6b,d). When Si degradation revealed the newly exposed surface during cycling, it not only continuously consumed the limited electrolyte and Li ions, but also increased the internal resistance of the cell by electrical contact loss and a thickened SEI layer. Because practical batteries use only highly limited electrolytes and Li sources, such degradation would severely limit the long-term cycle life. By contrast, the C(5)Si-G electrode sustains its morphological integrity, and the C(5)Si layer is well attached to the graphite (Fig. 6c,e; Supplementary Fig. 32 shows a photograph of the cycled electrode).

To clarify the preservation of the morphology of the C(5)Si layer, the crystallite structure of SiC was investigated using a ME-C(5) Si-C electrode during cycling. In this analysis, the reason we selected ME-C(5)Si-C is that both the amorphous phase of carbon and excessive SiC amount offer the advantage of detecting crystallite SiC. We observed that a clear peak assigned to SiC existed even after 500 cycles (half-cell) through the X-ray diffraction and TEM analyses (Fig. 6f,g), indicating that SiC with a high mechanical strength could act as a robust matrix to a Si volume change. A cross-sectional SEM image and magnified TEM images of ME-C(5)Si-C after 100 cycles are also shown in Supplementary Fig. 33. The C(5)Si layer does not show crack propagation at either the micro- or nanoscale, which means that there is no pathway or pores for the electrolyte to open up a crack and penetrate into the C(5)Si layer. We believe that the highly mitigated stress evolution originating from the decreased size of Si to the subnanometre level, along with the crystallite SiC and a-C matrices, preserves its morphological integrity and results in stable SEI layer formation without electrolyte penetration into the CSi layer (Fig. 6h for a schematic of the charge and discharge mechanisms of the pSi and CSi layer).

Beyond the small-sized stacked cell evaluation, performance verification in a practical-scale cell configuration is essential, particularly with the adoption of Si-based anodes, because the electrolyte depletion and safety issues caused by Si expansion become more severe in large-capacity cells. We fabricated a prismatic-type winding cell with a capacity above 110 Ah by using graphite-blended C(5) Si-G (C(5)Si-G/Gra) and a $\text{LiNi}_{0.8}\text{Co}_{0.1}\text{Mn}_{0.1}\text{O}_2$ (NCM811) cathode;

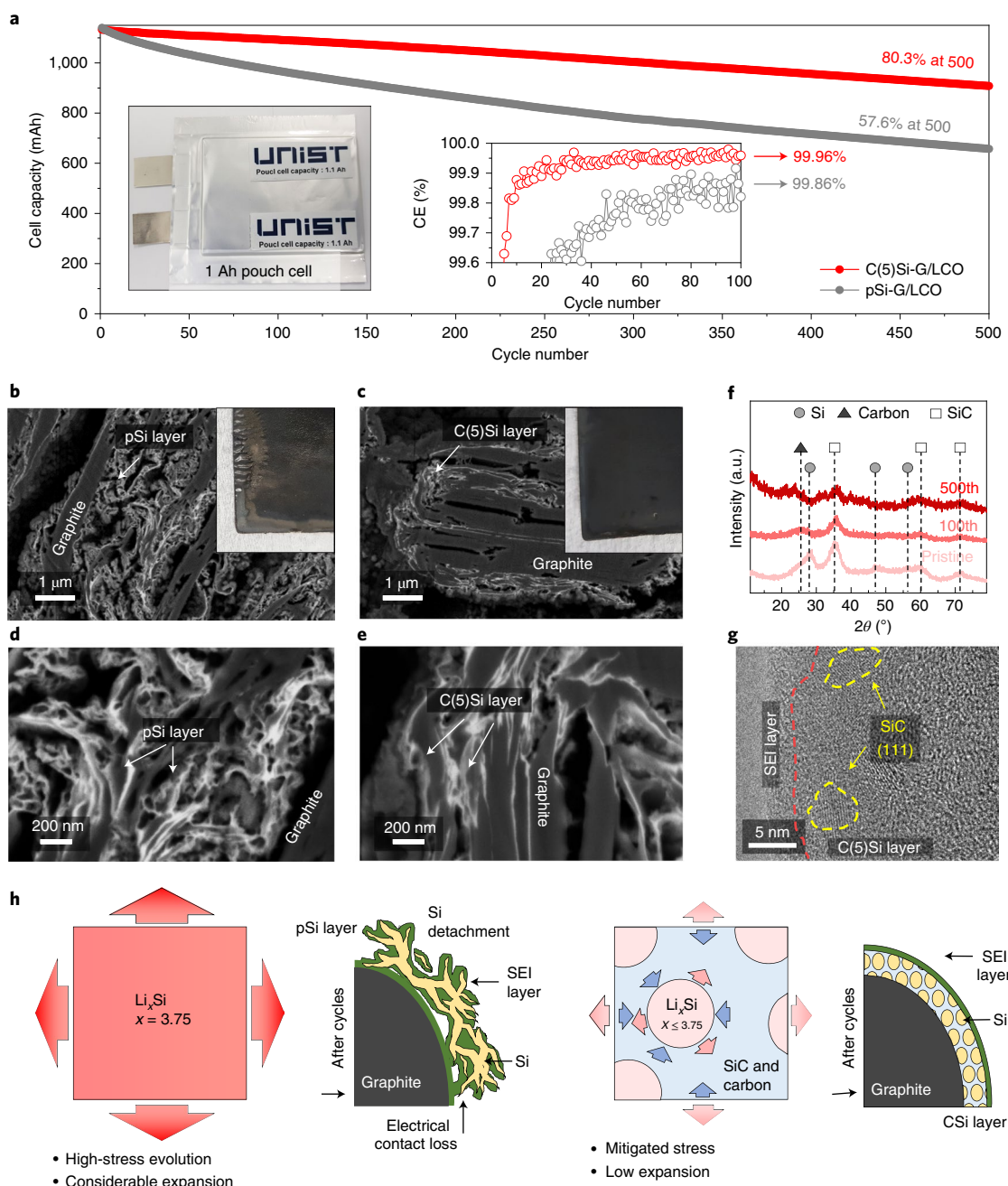


Fig. 6 | Behaviour of Si and Li_xSi in the CSi layer during cycles. **a**, Discharging capacity retention of pSi-G/LCO and C(5)Si-G/LCO when cycled at 1C (discharge) and 0.5C (charge) at 25 °C using 1Ah full-cell. Dotted line indicates the commercially acceptable limit. Insets: photograph and graph indicate a 1Ah pouch full-cell and CE, respectively. **b–e**, Cross-sectional SEM images of a pSi-G (**b,d**) and C(5)Si-G (**c,e**) electrode after the 500th cycle at the delithiation state with an inset showing a photograph of the top view of each electrode. All SEM images were obtained using a back-scattered electron mode to distinguish the carbon and Si clearly. Each experiment was repeated independently at least three times to obtain similar results. **f,g**, X-ray diffraction analysis of ME-C(5)Si-C depending on the number of cycles (pristine, 100 cycles, 500 cycles; **f**) and cross-sectional TEM image at 500 cycles (**g**). **h**, Schematic illustration of the behaviour of lithium silicide and corresponding fading mechanisms of pSi-G and C(5)Si-G.

Supplementary Fig. 34 shows the voltage profile and cycle stability with patterned cycling conditions. C(5)Si-G/Gra provided a surprisingly stable cycle lifespan (91.3%) for 2,875 cycles without any cell deformation (Fig. 7a). We also measured the in-situ cell thickness change of the C(5)Si-G/Gra during cycles in a 1 Ah pouch-type cell, as shown in the inset of Fig. 7a. The C(5)Si-G/Gra showed very low volume expansion (13.8%) at the first cycle, as well as a stabilized cell thickness change for repeated cycles. These low and stable

cell volume changes can avoid cell deformation and achieve stable electrochemical behaviour in densely packed prismatic cell designs.

To further confirm the safety of the 110 Ah prismatic-type cell, we conducted assorted safety testing such as external short circuit, thermal stability and overcharging (Fig. 7b–d; the detailed evaluation conditions are in the Methods), and these cells did not catch fire or burst. Because such a safe and prominent electrochemical performance gives us the promise of practical application, by using these

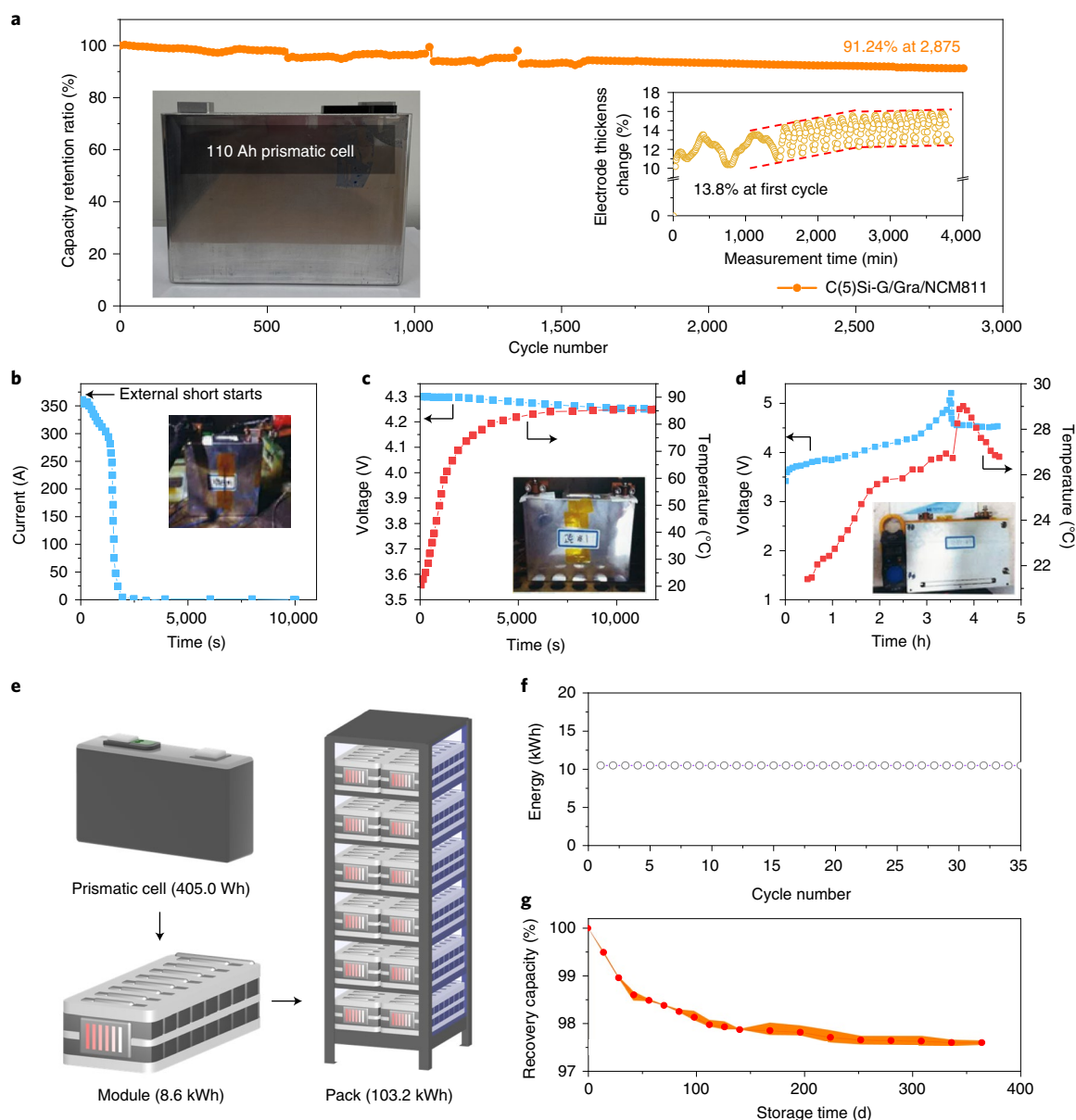


Fig. 7 | Practical application of subnano-sized Si. **a**, The 110 Ah prismatic-type full-cell using C(5)Si-G/Gra/NCM811. Insets: photographs of the 110 Ah prismatic-type full-cell and the electrode thickness change, which was measured by a 1 Ah pouch-type full-cell. **b–d**, Assorted safety tests: external short circuit (**b**), thermal stability (**c**) and overcharging (**d**). **e**, Schematic exhibiting a prismatic cell, module and pack for the fabrication of energy storage systems. **f**, Energy retention during energy storage system operation. **g**, Calendar life of 110 Ah prismatic-type cell (the orange shading shows the standard deviation calculated from three cells).

prismatic cells, we assembled a 8.6 kWh class battery module composed of 22 prismatic cells and then a 103.2 kWh prototype battery pack consisting of 12 battery modules as shown in Fig. 7e (detailed electrode composition and specifications in Supplementary Tables 6 and 7). Because this type of battery pack is a practically used model for energy storage systems, we used it to set up an energy storage system (Supplementary Fig. 35). When it operated in the voltage range between approximately 943 and 962 V, it exhibited an energy of 10.5 kWh without capacity fading or drastic temperature change (Fig. 7f and Supplementary Fig. 36). We also conducted a cell storage test—with details provided in the Methods—to measure the battery's calendar life (Fig. 7g). In this test, we traced the battery's recovery capacity for a full year and found a 97.6% recovery capacity after 365 days.

Conclusions

We synthesized subnano-sized Si anodes using ethylene as a particle growth inhibitor in a thermal decomposition process of silane. In contrast to the decomposition of pure silane, where the Si particles continue to increase in size, ethylene reacts with silane, forming Si–C bonds, thereby maintaining the subnano-sized particles even when an excessive coating is present on the carbonaceous substrate. Moreover, forming Si–C bonds contributes to generating the robust matrix of crystallite SiC, which not only prevents the formation of a native SiO₂ layer but also helps to preserve the morphological integrity of the C(5)Si layer during long-term cycling. Thus, C(5)Si-G demonstrates a high initial CE and cycling stability when estimated using a practically feasible testing protocol. The fabricated energy storage system (107 kWh) consisting of 110 Ah full-cells contain-

ing subnano-sized Si anodes showed cycling stability (91% for 2,875 cycles) and a good calendar life (97.6% for 365 days). These results can provide inspiration for battery researchers to consider the parameters for high-capacity alloy anode design and can pave the way for next-generation anodes in high-energy lithium-ion battery systems for application in energy storage systems and electric vehicles.

Methods

Synthesis of CSi-G. The Si layer was coated on the substrate (commercial spherical-type natural graphite and carbon nanoparticle) in a customized stainless steel tube furnace via thermal decomposition of high-purity monosilane gas (99.9999%) at 475 °C at a flow rate of 0.05 l min⁻¹ for 35 and 60 min (denoted Si-G and E-Si-G respectively) using a graphite substrate. For the fabrication of the C(5) Si layer, high-purity monosilane and ethylene (99.9%) were streamed at 475 °C at a flow rate of 0.05 l min⁻¹ at a volume ratio of 10:5 for 45 min (denoted C(5)Si-G), 78 min (denoted E-C(5)Si-G) and 165 min (denoted ME-C(5)Si-C, where C is the amorphous carbon nanoparticle substrate (Super P, TIMCAL)), and at 10:1.5 for 40 min (denoted C(1.5)Si-G). All samples were coated by 5 wt% pitch-based carbon (10 wt% carbon coat for the ME-C(5)Si-C anode) and annealed at 900 °C in the case of electrochemical characterization. As the total purity of the gas mixture of monosilane and ethylene was not as high as 99.9999%, utilization of a highly purified gas mixture is not essential for the synthesis of CSi-G.

Material characterization. Structural investigation of the CSi layer was conducted using SEM (Verios 460, FEI) and high-resolution TEM (JEM-2100F, JEOL) with EDS (Aztec, Oxford). For the TEM analysis, dual-beam focused ion beam milling (Helios 450HP, FEI) with a 2–30 kV Ga ion beam was used for the preparation of samples. Before analysis of the focused ion beam milling, epoxy soaking and carbon deposition were performed to avoid damage from the Ga ion beam and maintain the sample morphology in the process of lift-out and thinning. The thinned samples with a thickness of 200 nm were next polished to remove surface contamination using an Ar-ion milling system (Model 1040 Nanomill, Fischione) and immediately transferred to the TEM instrument. The chemical states of each layer were analysed by EELS and XPS (K-alpha, ThermoFisher). EELS data were calibrated against the π^* peak of the K edge of amorphous carbon, which is accurately locked at 285 eV. For the calibration of the binding energy of the XPS spectra, we used the adventitious carbon 1s peak of 284.5 eV as the internal standard. An X-ray diffractometer (D/MAXZ 2500V/PC, Rigaku, with Cu K α of wavelength 1.542 Å) and Raman spectroscopy (NRS-5000, JASCO) were used for the physicochemical characterization. The elemental composition of C(5)Si-G was analysed by element analyser (Flash 2000, Thermo). To examine the morphological change of electrodes and samples after 500 cycles, the cycled pouch-type cells were disassembled in an Ar-filled glove box (H₂O < 1 ppm, O₂ < 1 ppm), and electrodes were rinsed with excess dimethyl carbonate. Then, the electrode was cut by an ion milling system (IM-40000, Hitachi) for analysis of cross-sectional SEM images and cut by focused ion beam milling (above-mentioned procedures) for analysis of cross-sectional TEM images. The ionic conductivity of the electrolyte was measured by a conductivity meter (Eutech CON 150, ThermoFisher). The cyclic voltammetry curve was measured by a Wonatech WBCS3000 at a scan rate of 0.5 mV s⁻¹.

Dilatometry. The thickness change of the 1 Ah pouch-type full-cell (C(5)Si-G/LCO and pSi-G/LCO; LCO, lithium cobalt oxide) after 20 cycles was estimated by an electrochemical dilatometer (Mitutoyo). Electrochemical conditions were the same as those in the electrochemical characterization of the full-cell except for temperature (45 °C). In the case of the 1 Ah pouch-type full-cell (C(5)Si-G/Gra/NCM811), the measurement of electrode thickness changes was performed after a degassing process.

Electrochemical characterization of a half-cell. The electrode was fabricated using the active material (Si-G, E-Si-G, C(1.5)Si-G, C(5)Si-G and E-C(5)Si-G), the carbon black (Super P, TIMCAL), carboxymethyl cellulose (Nippon paper) and styrene-butadiene rubber (Zeon) at the mass ratio of 96:1:1.5:1.5. The slurry was cast on copper foil up to 5.35 mg cm⁻² (Si-G, C(1.5)Si-G and C(5)Si-G) and 3.3 mg cm⁻² (E-Si-G and E-C(5)Si-G). All electrodes were then dried at 110 °C for 6 h in the vacuum oven and calendared to increase the electrode density to 1.6 g cm⁻³. A CR2032 coin-type half-cell was assembled by sandwiching the microporous polyethylene (20 μ m, Celgard) between the electrodes and lithium metal foil (>99%, Honjo Metal) in the Ar-filled glove box. Then 1.3 M LiPF₆ in a mixture of ethylene carbonate, ethyl methyl carbonate and diethyl carbonate (3:5:2, v/v) with 10% fluoroethylene carbonate, 0.2% lithium tetrafluoroborate (LiBF₄), 0.5% vinylene carbonate, 3% succinonitrile and 1% propane sultone (Panax Starlyte) was used as the electrolyte. The electrochemical assessment was carried out in a voltage range of 0.005 V (cut-off voltage, 0.01 V) to 1.5 V at a rate of 0.1 C for the first cycle, and 0.005 V (constant voltage mode for charging, where the cut-off voltage was 0.01 C) to 1.0 V at a rate of 0.5 C for the rest of the cycles.

All cells were charged at constant current and CV mode for subsequent cycles. The LCO electrode was fabricated by mixing LCO powder, Super P and polyvinylidene fluoride binder with a mass ratio of 96:2:2 on the aluminium foil. The mass loading level and electrode density of the LCO were 15.5 mg cm⁻² and 3.5 g cm⁻³, respectively. The separator, electrolyte and battery cyclers were used in the same way as in the above anode tests. The electrochemical evaluations were assessed in a voltage range between 3 and 4.2 V at a rate of 0.1 C at the first cycle and 0.5 C for the rest of the cycles. Electrochemical analysis of all coin-type half-cells was conducted using a battery cycler (TOSCAT-3100, TOYO SYSTEM) at 25 °C.

Electrochemical characterization of full-cell. The electrode was punched into rectangular pieces with a size of 72.5 mm $\times$ 55.0 mm (cathodes) and 75.5 mm $\times$ 58.0 mm (anodes) for the pouch-type full-cells with >1,000 mAh. The pouch-cell assembly was conducted with stacking of the cathode, anode and separator at ~7–8 layer construction; ultrasonic welding for connection of the electrode with Al (cathode) and Ni (anode) external tabs; and vacuum sealing with electrolyte injection in a dry room (relative humidity, 0.05%; 20 °C). The electrode composition of the cathode or anode, separator and electrolyte was the same as those of half-cells except that a slurry was coated on both sides of a current collector such as copper and aluminium. Both the areal capacity ratio of negative to positive electrode and ratio of electrolyte weight to cell capacity are <1.1 and <2 g Ah⁻¹, respectively, with the limited Li source and electrolyte, to correspond to the values used in conventional lithium-ion batteries. The electrochemical evaluations were assessed in a voltage range between 2.5 and 4.2 V at a rate of 0.1 C at the first cycle and 0.2 C for three cycles (ageing and degassing processes were conducted at the first cycle). For the rest of the cycles, cells were charged and discharged in a voltage range between 2.85 and 4.2 V at a rate of 0.5 C. Electrochemical analysis of all pouch-type full-cells was carried out using a battery cycler (PECC05-06, PNE SOLUTION) at 25 and 45 °C. For the case of the 110 Ah prismatic-type full-cell, the electrode was fabricated using an anode (C(5)Si-G/Gra; loading level, 7.5 mg cm⁻²) and cathode (NCM811; loading level, 13.4 mg cm⁻²), and the electrode composition was the same as with the C(5)Si-G electrode. The pressure of the 1 Ah pouch-type full-cell and 110 Ah prismatic-type full-cell was 5 and 10 psi, respectively. All electrochemical tests in this study were conducted in isothermal chambers with a constant temperature. In the case of the pack-scale system for energy storage systems, the system also had an active cooling fan to avoid the overheating of the system. The electrochemical evaluations (cycle test) were assessed under a patterned cycle condition used for practical energy storage system operation. In the testing of the calendar life, the cell was stored in a state of charge of 25% at 30 °C. The phenomena of knee-type drop-off and the electrolyte drying out were not considered when expecting the calendar life using the trend curve. The fabrication was supported by Samsung SDI, and the other evaluation conditions and fabrication processes are confidential.

Safety testing of 110 Ah prismatic full-cell. For the external short-circuit test, the cell was fully charged with a state of charge of 100% and then rested for 12 h. Then, an external short circuit was conducted by connecting the anode and cathode at an external resistance of approximately 30 m Ω for 6 h. For the thermal test, a fully charged cell was heated to 85 °C at a heating rate of 5 °C min⁻¹ and held there for 3 h. For the overcharging test, after reaching a fully discharged state (depth of discharge, 100%), the cell rests for 1 h; then, we started to charge the cell at 0.33 C until a cut-off voltage was reached of 5.22 V (overcharging to 120% of the initial voltage).

Molecular simulations. *DFT calculations.* Spin-polarized DFT calculations were performed using the Gaussian 16 program³⁹. The Becke's three-parameter non-local exchange⁴⁰ and Lee–Yang–Parr non-local correlation⁴¹ (B3LYP) functional were employed with the 6-31+G(d,p) basis set. To understand the thermal decomposition pathway of the cogas, we estimated the bond dissociation energy (BDE) for the reactants and heats of reaction (ΔE) for plausible elementary reaction steps, which could be observed in the decomposition reaction. The BDE for each bond in the reactants was calculated as follows:

$$\text{BDE} = -(E_{\text{elong}} - E_0) \quad (1)$$

where E_0 is the total energy of the optimized molecule and E_{elong} is the total energy of the molecule whose bond is elongated by 4 Å.

Reax-MD simulations. All reax-MD simulations were conducted with the reactive forcefield (reaxFF) developed by Soria et al.⁴² and implemented in the Large-scale Atomic/Molecular Massively Parallel Simulator (LAMMPS) package^{43,44}. Reax-MD simulations were carried out in a two-step approach: formation of a Si cluster from precursor gases (Supplementary Fig. 3a) and formation of a CSi layer from the nucleated Si clusters (Supplementary Fig. 3b). To mimic the experimental environment, we modelled a graphene-mixture gas (SiH₄ and C₂H₄) interfacial system ('Initial system I' in Supplementary Fig. 3a), which consisted of three layers of graphene and a mixture of monosilane and ethylene with a density of 0.05 g cm⁻³. In this study, three ratios of monosilane and ethylene were considered (that is, 10:0, 10:1.5 and 10:5, denoted Si, C(1.5)Si and C(5)Si, respectively), and

the number of monosilane molecules was fixed at 159 (Supplementary Table 2). The interfacial model systems were relaxed through a *NVT* (that is, *N*, *V* and *T* are the total number of atoms, volume and temperature at each simulation condition) reax-MD simulation at room temperature for 100 ps with a time step of 0.1 fs. After that, the simulation temperature was elevated to the reaction temperature (that is, 1,800 K) at a heating rate of 250 K ps⁻¹. Thermal decomposition reactions of the heated model systems were then propagated through *NVT* reax-MD simulation for 1 ns. Pressurized conditions of the mixture gas precursors and elevated reaction temperature (that is, 0.5 g cm⁻³ and 1,800 K) were chosen to allow the decomposition reactions of the mixture gas to be completed in a practical simulation time. Similar approaches have been applied in several previous studies observing reactions of SiC (refs. ^{45,46}).

To observe the structure of the CSi layer with and without ethylene gas, Si clusters grown in the Si and C(1.5)Si systems were taken from the last configuration of reax-MD. Ten copies of the Si clusters were randomly packed in an empty simulation box with a packing density of 0.5 g cm⁻³ (Initial system II' in Supplementary Fig. 3b). Note that two small Si clusters were made from the complex gas system, so ten copies of each cluster (a total of 20 clusters) were packed in the Initial system II of the C(1.5)Si system. Modelled systems were quenched from the simulation temperature (1,800 K) to the experimental reaction temperature (900 K) with a cooling rate of 250 K ps⁻¹, and a further *NPT* (isothermal and isobaric, *P* is the pressure of the simulation system) reax-MD simulation was conducted at 50 atm for 200 ps.

Data availability

Data generated and analysed in this study are included in the article and its Supplementary Information.

Received: 25 March 2021; Accepted: 20 October 2021;

Published online: 13 December 2021

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Acknowledgements

This work was supported by the Korea Institute of Energy Technology Evaluation and Planning and the Ministry of Trade, Industry and Energy of the Republic of Korea (no. 20172410100140). This research was also supported by the Basic Science Research Program through the National Research Foundation of Korea, funded by the Ministry of Education (2019R1A6A3A13095900). Fabrication of the 1 Ah and 110 Ah cells was supported by K. Kim and Samsung SDI.

Author contributions

J.S., J.M. and N.K. conceived and designed the experiments. J.S., J.M. and N.K. prepared the samples and carried out the main experiments. J.H.L., S.H.J. and S.K.K.

performed the simulation using MD and DFT. S.C. conducted dilatometry. M.Y. and J.H. participated in electrochemical measurements. T.L. and Y.L. assisted with sample preparation. J.S., N.K. and J.H.L. wrote the paper. J.C., N.K. and S.K.K. discussed the results and revised or commented on the manuscript.

Competing interests

The authors have patents (J.C., J.S. and J.M., Patent Cooperation Treaty patent publication no. WO/2020/036397, published 20 February 2020. J.C., J.S. and J.M., Korean patent application no. 1021312620000, registered 1 July 2020. J.C., J.S. and J.M., Korean patent application no. 1022130820000, registered 1 February 2021. J.C. and J.S., Korean patent application no. 1021596930000, registered 18 September 2020.) related to the processes described in this article.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41560-021-00945-z>.

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Peer review information *Nature Energy* thanks Taylor Garrick and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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